

Smooth Manifolds and Whitney's Embedding Theorem

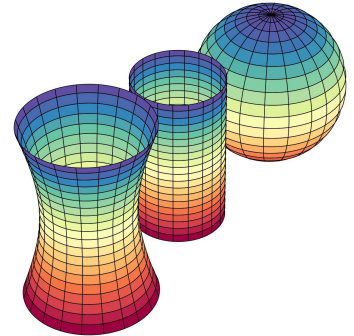
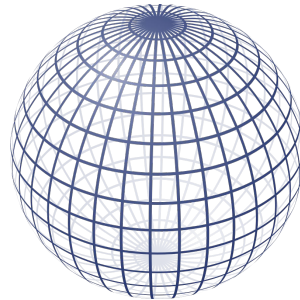
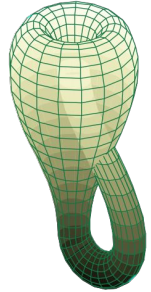
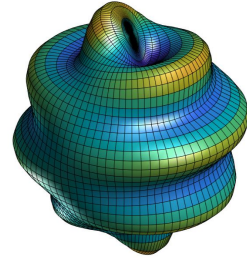
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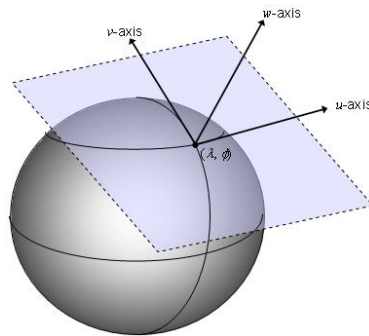
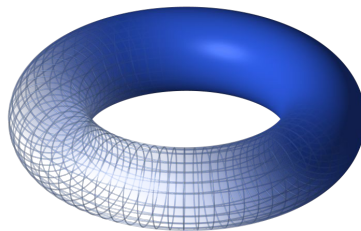
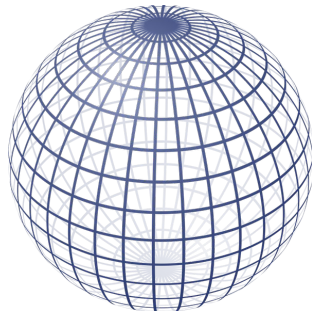
What is a smooth Manifold?

- Suppose that you are a very small object (ant) on a surface
- If, at every point on the surface, it appears to be “like” some Euclidean Space, then that surface is called a manifold
- A smooth manifold is one where there are no sharp corners, cusps, etc.



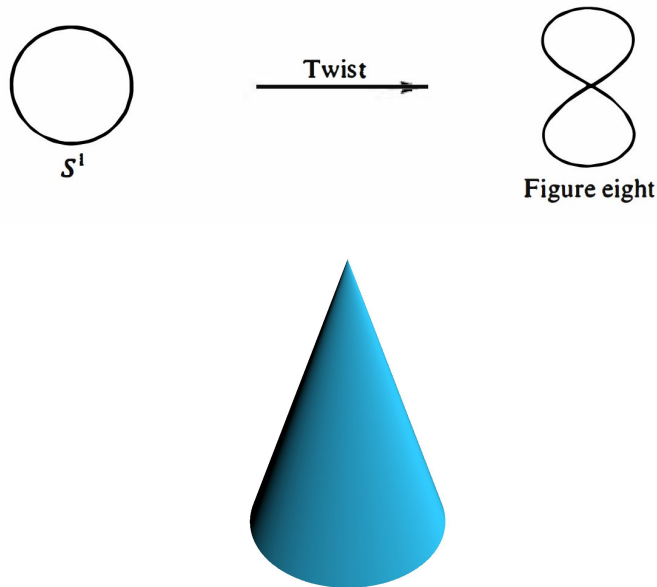
Examples

- Earth (S^2)
 - We are small objects on earth, and locally, a neighborhood looks like $\mathbb{R}^2$ near us
- Wire
 - Imagine you are an ant on a wire: to you, the wire will appear like $\mathbb{R}$
- Donut (torus)
 - Different from the sphere, yet another $\mathbb{R}^2$ manifold
- Euclidean Space



Non-examples

- Figure Eight in $\mathbb{R}^2$
 - Every point is locally “like” $\mathbb{R}$, except where the figure eight intersects with itself
- Cube, rectangular prism and cone in $\mathbb{R}^3$
 - Most of the points on these surfaces are “like” $\mathbb{R}^2$, except only a few points



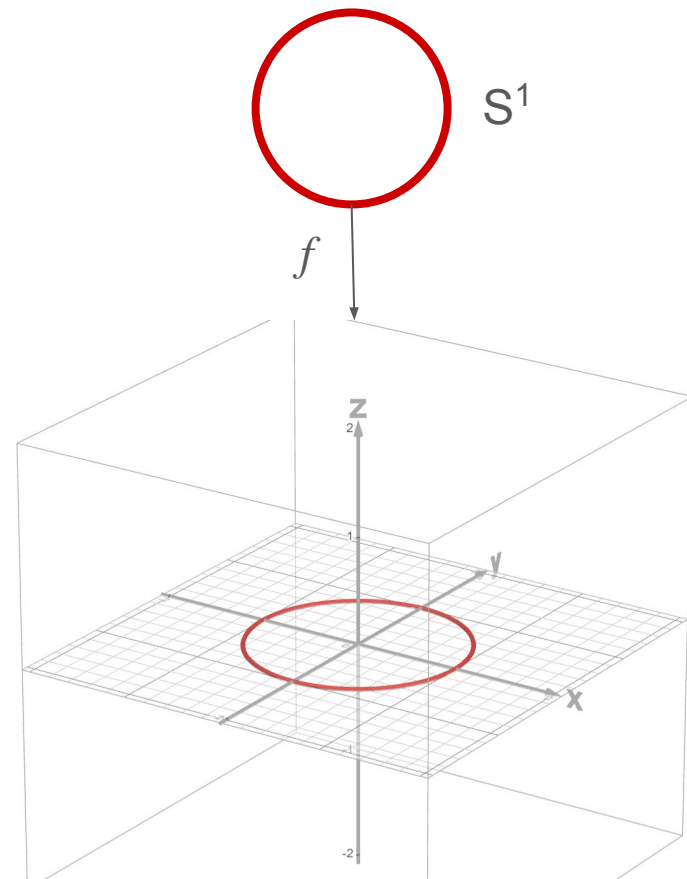
Purpose and Uses of Smooth Manifolds

- Calculus
 - Because there are no sudden corners, etc., we can take limits, derivatives, integrals and do other calculus operations
 - Have a wide range of applications from physics to engineering
- Place them in Euclidean Space

What is an embedding?

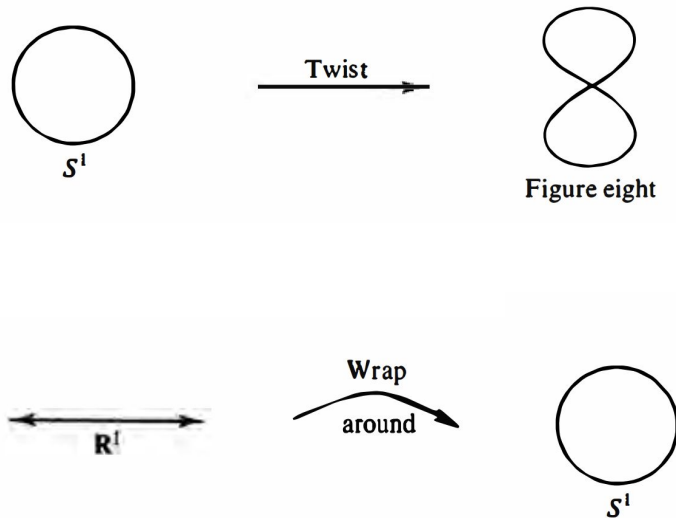
Immersions

- Involves placing “small” manifolds in “larger” manifolds
- Consider $f: X \rightarrow \mathbb{R}^n$, where $\dim X = k < n$, X a manifold
 - Place [local] coordinates on $x \in X$ and $y = f(x) \in \mathbb{R}^n$
 - If there are [local] coordinates such that $f(x_1, x_2, \dots, x_k) = (x_1, x_2, \dots, x_k, 0, 0, \dots, 0)$, then f is said to be an immersion at x
 - This is a local characteristic, not a global one
- For example: $f: S^1 \rightarrow \mathbb{R}^3$
 - Takes a circle, which lives in $\mathbb{R}^2$, and puts it in $\mathbb{R}^3$



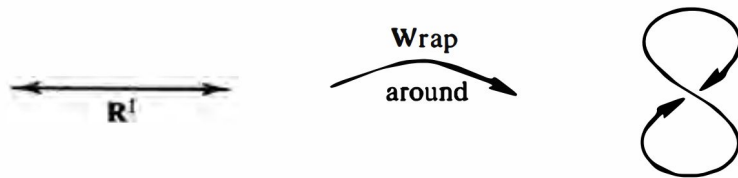
Immersions (cont.)

- Although immersions are powerful, they do not guarantee the structure of the manifold remains intact
 - The twisting map shown is an immersion, but its image is not even a manifold
 - $f: \mathbb{R} \rightarrow \mathbb{R}^2, f(t)=(\cos(t), \sin(t))$ is locally an immersion, but it compresses the entire real line down to a circle
 - Not injective as $f(t)=f(t+2\pi)$
 - $\mathbb{R}$ is not compact, but S^1 is, so even compactness of the range is not enough!



Immersions and Embeddings

- Is injectivity good enough? No!
- We need one more condition:
proper
 - the preimages of compact sets are compact
- Putting the injective, proper, and immersion conditions together, we get embeddings
 - Embeddings preserve *everything* about the manifold, while allowing you to treat it as a submanifold of the larger space
- How big does the ambient Euclidean Space need to be?

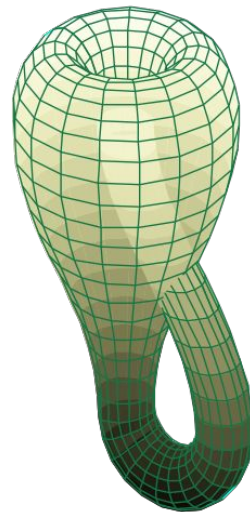


Whitney's Embedding Theorem

How much room do we need?

- If X is a k -dimensional manifold, how much room do we need to embed it in Euclidean space?
- S^1 doesn't embed in $\mathbb{R}$, but it does in $\mathbb{R}^2$
- S^2 doesn't embed in $\mathbb{R}^2$, but it does in $\mathbb{R}^3$
 - Are we on to something? Is $k+1$ enough?
- No! The Klein bottle is a 2-dimensional manifold...

but it has self-intersections in $\mathbb{R}^3$!



Weak and Strong Whitney Embedding Theorems

- First, he came up with $n \geq 2k+1$
 - Weak Embedding Theorem
- After some work, he reduced it to $n \geq 2k$
 - Strong Embedding Theorem
- So we can say confidently that the Klein Bottle can be embedded in $\mathbb{R}^4$!

Exploring the Weak Whitney Embedding Theorem

- Consider the example to the right
 - A 1-dimensional, self intersecting curve is shown
 - Embedding this in a larger space, we can remove the intersection and create a manifold
- Two pieces of 1D objects can be separated in 3D space
- Similarly, pieces of 2D objects can be separated in 5D space
- For a k -dimensional manifold, we can embed in the $2k+1$ Euclidean space remove self-intersections



Thanks for Listening!